

Table 3.1 – Values of  $k_{mod}$ 

| Material               | Standard                                                       | Service class | Load-duration class |                  |                    |                   |                      |
|------------------------|----------------------------------------------------------------|---------------|---------------------|------------------|--------------------|-------------------|----------------------|
|                        |                                                                |               | Permanent action    | Long term action | Medium term action | Short term action | Instantaneous action |
| Solid timber           | EN 14081-1                                                     | 1             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                | 2             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                | 3             | 0,50                | 0,55             | 0,65               | 0,70              | 0,90                 |
| Glued laminated timber | EN 14080                                                       | 1             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                | 2             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                | 3             | 0,50                | 0,55             | 0,65               | 0,70              | 0,90                 |
| LVL                    | EN 14374, EN 14279                                             | 1             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                | 2             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                | 3             | 0,50                | 0,55             | 0,65               | 0,70              | 0,90                 |
| Plywood                | EN 636<br>Part 1, Part 2, Part 3<br>Part 2, Part 3<br>Part 3   | 1             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                | 2             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                | 3             | 0,50                | 0,55             | 0,65               | 0,70              | 0,90                 |
| OSB                    | EN 300<br>OSB/2<br>OSB/3, OSB/4<br>OSB/3, OSB/4                | 1             | 0,30                | 0,45             | 0,65               | 0,85              | 1,10                 |
|                        |                                                                | 1             | 0,40                | 0,50             | 0,70               | 0,90              | 1,10                 |
|                        |                                                                | 2             | 0,30                | 0,40             | 0,55               | 0,70              | 0,90                 |
| Particle-board         | EN 312<br>Part 4, Part 5<br>Part 5<br>Part 6, Part 7<br>Part 7 | 1             | 0,30                | 0,45             | 0,65               | 0,85              | 1,10                 |
|                        |                                                                | 2             | 0,20                | 0,30             | 0,45               | 0,60              | 0,80                 |
|                        |                                                                | 1             | 0,40                | 0,50             | 0,70               | 0,90              | 1,10                 |
|                        |                                                                | 2             | 0,30                | 0,40             | 0,55               | 0,70              | 0,90                 |
| Fibreboard, hard       | EN 622-2<br>HB.LA, HB.HLA 1 or 2<br>HB.HLA1 or 2               | 1             | 0,30                | 0,45             | 0,65               | 0,85              | 1,10                 |
|                        |                                                                | 2             | 0,20                | 0,30             | 0,45               | 0,60              | 0,80                 |
| Fibreboard, medium     | EN 622-3<br>MBH.LA1 or 2<br>MBH.HLS1 or 2<br>MBH.HLS1 or 2     | 1             | 0,20                | 0,40             | 0,60               | 0,80              | 1,10                 |
|                        |                                                                | 1             | 0,20                | 0,40             | 0,60               | 0,80              | 1,10                 |
|                        |                                                                | 2             | –                   | –                | –                  | 0,45              | 0,80                 |
| Fibreboard, MDF        | EN 622-5<br>MDF.LA, MDF.HLS<br>MDF.HLS                         | 1             | 0,20                | 0,40             | 0,60               | 0,80              | 1,10                 |
|                        |                                                                | 2             | –                   | –                | –                  | 0,45              | 0,80                 |

(4) For timber which is installed at or near its fibre saturation point, and which is likely to dry out under load, the values of  $k_{def}$ , given in Table 3.2, should be increased by 1,0.

(5)P Finger joints shall comply with EN 385.

### 3.3 Glued laminated timber

(1)P Glued laminated timber members shall comply with EN 14080.

NOTE: In EN 1194 values of strength and stiffness properties are given for glued laminated timber allocated to strength classes, see annex D (Informative).

(2) The effect of member size on strength may be taken into account.

(3) For rectangular glued laminated timber, the reference depth in bending or width in tension is 600 mm. For depths in bending or widths in tension of glued laminated timber less than 600 mm

### 3.2 Solid timber

Delete paragraph (1)P and replace with:

(1)P Timber members shall comply with EN 14081-1.

NOTE: Strength classes for timber are given in EN 338.

#### 3.1.3 Strength modification factors for service classes and load-duration classes

Table 3.1, delete and replace with

Table 3.1 – Values of  $k_{mod}$

| Material               | Standard                                                             | Service class | Load-duration class |                  |                    |                   |                      |
|------------------------|----------------------------------------------------------------------|---------------|---------------------|------------------|--------------------|-------------------|----------------------|
|                        |                                                                      |               | Permanent action    | Long term action | Medium term action | Short term action | Instantaneous action |
| Solid timber           | EN 14081-1                                                           | 1             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                      | 2             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                      | 3             | 0,50                | 0,55             | 0,65               | 0,70              | 0,90                 |
| Glued laminated timber | EN 14080                                                             | 1             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                      | 2             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                      | 3             | 0,50                | 0,55             | 0,65               | 0,70              | 0,90                 |
| LVL                    | EN 14374, EN 14279                                                   | 1             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                      | 2             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                      | 3             | 0,50                | 0,55             | 0,65               | 0,70              | 0,90                 |
| Plywood                | EN 636<br>Type EN 636-1<br>Type EN 636-2<br>Type EN 636-3            | 1             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                      | 2             | 0,60                | 0,70             | 0,80               | 0,90              | 1,10                 |
|                        |                                                                      | 3             | 0,50                | 0,55             | 0,65               | 0,70              | 0,90                 |
| OSB                    | EN 300<br>OSB/2<br>OSB/3, OSB/4<br>OSB/3, OSB/4                      | 1             | 0,30                | 0,45             | 0,65               | 0,85              | 1,10                 |
|                        |                                                                      | 1             | 0,40                | 0,50             | 0,70               | 0,90              | 1,10                 |
|                        |                                                                      | 2             | 0,30                | 0,40             | 0,55               | 0,70              | 0,90                 |
| Particle-board         | EN 312<br>Type P4, Type P5<br>Type P5<br>Type P6, Type P7<br>Type P7 | 1             | 0,30                | 0,45             | 0,65               | 0,85              | 1,10                 |
|                        |                                                                      | 2             | 0,20                | 0,30             | 0,45               | 0,60              | 0,80                 |
|                        |                                                                      | 1             | 0,40                | 0,50             | 0,70               | 0,90              | 1,10                 |
|                        |                                                                      | 2             | 0,30                | 0,40             | 0,55               | 0,70              | 0,90                 |
| Fibreboard, hard       | EN 622-2<br>HB.LA, HB.HLA 1 or 2<br>HB.HLA1 or 2                     | 1             | 0,30                | 0,45             | 0,65               | 0,85              | 1,10                 |
|                        |                                                                      | 2             | 0,20                | 0,30             | 0,45               | 0,60              | 0,80                 |
| Fibreboard, medium     | EN 622-3<br>MBH.LA1 or 2<br>MBH.HLS1 or 2<br>MBH.HLS1 or 2           | 1             | 0,20                | 0,40             | 0,60               | 0,80              | 1,10                 |
|                        |                                                                      | 1             | 0,20                | 0,40             | 0,60               | 0,80              | 1,10                 |
|                        |                                                                      | 2             | –                   | –                | –                  | 0,45              | 0,80                 |
| Fibreboard, MDF        | EN 622-5<br>MDF.LA, MDF.HLS<br>MDF.HLS                               | 1             | 0,20                | 0,40             | 0,60               | 0,80              | 1,10                 |
|                        |                                                                      | 2             | –                   | –                | –                  | 0,45              | 0,80                 |